Qualification Programme Examination Panelist's Report

Module 3 – Business Economics
(June 2022 Session)

(The main purpose of the following report is to summarise candidates' common weaknesses and make recommendations to help future candidates improve their performance in the examination.)

(I) Section A – Multiple Choice Questions

General Comments

This section consisted of 15 multiple-choice questions covering the whole syllabus (on economics and business statistics). Good performance in this section would reflect that candidates had developed a solid understanding of the relevant concepts and the ability to apply the concepts in different settings as well as handle the calculations well.

Overall, candidates performed well in the category of "Knowledge" but demonstrated marginally weaker performance under "Application" and "Computation".

Specific Comments

Candidates' performance was weak in the following areas:

- Applying basic definitions to determine the outcome in scenarios that involve several possible outcomes (e.g., MC4 ("knowledge" and "application" categories) – How does a notebook computer whose price is twice that of a desktop computer contribute to GDP? The key point to note here is that both types of computers would be under the same category in the GDP calculation and thus their prices would determine their relative contributions to the output measure.)

- Unfamiliarity with standard but key concepts and formulas and their use in both economic and statistical issues (e.g., MC11 ("calculation" category) – mid-point price elasticity of demand calculation, MC14 ("calculation" category) – standard error of sample mean, and MC5 ("knowledge" category) – definition of a perfectly inelastic supply curve)

In preparing for the exam, candidates are advised to review the issues from the three perspectives mentioned above (i.e., knowledge, calculation, and applications).

Moreover, an efficient and effective way for candidates to gain a basic understanding of the whole syllabus and the types of questions that they might face in the examination is to look at past exam papers and review the relevant sections of the supplementary readings, including the economics and statistics reference books assigned for this module.

(II) Section B – Written Questions

General Comments

This section consisted of four questions. Performance ranged from unsatisfactory to good, with an overwhelming majority skewing to the former.

Specific Comments

Question 1(a) – 2 Marks

The question asked candidates to state the general formula for the expenditure approach used in measuring the gross domestic product (GDP) of an open economy.

Overall performance was good. The overwhelming majority of candidates were able to answer this definition-based question correctly.

Question 1(b)(i) – 5 Marks

This question required candidates to apply the concept and formula for the spending multiplier in a closed and no-tax (hypothetical) economy to determine the values of the multiplier, marginal propensity to consume (MPC), and marginal propensity to save (MPS).

Overall performance was highly unsatisfactory. Most candidates, even if they seemed to know the basic multiplier formula, were not able to rearrange the equation to determine the multiplier's value. This failure likely reflected the more general problem associated with learning in a "copy-and-paste" mode that tended to obstruct learners' ability to apply the concept (underlying the formula/math) even in slightly different settings. A fundamental way to overcome this issue is to sufficiently understand the concept that related the variables (MPC and multiplier) in the formula. It is important for candidates to remember that economics is a broad subject wherein its associated mathematical techniques are simply supplementary tools.

Question 1(b)(ii) – 5 Marks

This question continued with part (b)(i) and was intended to test candidates' understanding of how the multiplier effects work. To help candidates organize their thinking about the concept behind the numbers involved in a multiplier process, the answers were to be presented in a fill-in-the-table format.

Overall performance was even more unsatisfactory than in the previous part. The poor performance seemed to confirm the observation made in the previous part that most of the candidates did not have a clear enough understanding of the "story" (concept) regarding how the multiplier process worked.

Candidates are strongly advised to consult the economics reference books when preparing for the examination.

Question 1(b)(iii) – 3 Marks

This question was a concept-based question and could be answered independently of the two previous parts. Specifically, this question tested candidates' understanding of how an increase in MPC and MPS would affect the multiplier.

Overall, performance was as unsatisfactory as in the two previous parts. Most candidates failed to understand the channel through which an initial change in spending (e.g., an increase in investment in the economy) would get "passed down" the "income and spending chain" of the economy, and in the process, how the size of MPC would affect how much income received by one sector would be spent in another sector.

The idea of the multiplier process can be represented quite clearly by the following general formula:

Total increase in spending = i + (i.MPC) + (i.MPC).MPC + [(i.MPC).MPC].MPC + …

where i is the amount of the initial spending.

It can be seen clearly from the formula above that the size of "total increase in spending" is positively related to MPC. As MPC = 1 – MPS, it can also be concluded that "total increase in spending" is negatively related to MPS.

Note that a more detailed discussion can be found in the economics reference books mentioned at the end of Section A above.

Question 1(b)(iv) – 4 Marks

This question required candidates to provide any two factors that would affect MPC.

Overall performance was about the same as the previous parts. Most of the candidates were simply able to "state" one correct factor with no discussion of how the factor affects MPC.

Question 1(b)(v) – 3 Marks

This question required candidates to analyse how the multiplier effect would be affected if the "closed and no-tax economy" assumptions were dropped.

Overall performance was about the same as in the previous parts. Of all parts that make up Question 1(b), this was the most difficult one. The key point candidates needed to note was: In a closed and no-tax economy, the only "leakage" that would reduce the multiplier effect is saving (which, as discussed in part (b)(iii) above, would reduce MPC). In an open and with-tax economy, two other leakages would work to reduce the multiplier effect: income tax reduces disposable/take-home income, while imports decrease the amount spent on domestic goods in every subsequent round of spending. Mathematically, the open economy multiplier is equal to 1 / (1-MPC+MPM), where MPM is the marginal propensity to import.

Question 2(a) – 5 Marks

This question required candidates to analyse a scenario in which a decrease in the market supply of a commodity (with all else the same) would result in a shortage of the commodity.

Overall, performance was satisfactory. The key point to keep in mind here was that shortage or surplus would only result if the market price was somehow prevented from adjusting, usually resulting from some external force such as price regulation by the government.

Question 2(b)(i) – 5 Marks

This question required candidates to calculate the new monthly dollar sales after a price change based on the estimated price elasticity of demand.

Overall performance was satisfactory. As the question involved rather straightforward calculations and was asked in a standard fashion, performance should have been much better. This reflects a lack of preparation and the candidates' weak background in economics.

Question 2(b)(ii) – 3 Marks

This question required candidates to estimate the (expected) change in revenue resulting from the proposed price change.

Overall performance was unexpectedly unsatisfactory. Given that the question was a straightforward calculations-based question, the poor performance once again pointed to candidates' lack of preparation and weak economics background. Consulting an economics reference book is strongly recommended as part of candidates' preparation for the exam.

Question 2(b)(iii) – 3 Marks

This question required candidates to apply the definition of unit elastic demand to determine whether a price change would result in a change in dollar sales (total revenue).

Overall performance was unsatisfactory. As this question was essentially a definition-based question and involved no calculations, this poor performance was likely due to a lack of preparation and candidates' weak background in economics.

Question 2(c)(i) – 3 Marks

This question required candidates to explain how a negative demand shock (due to bird flu) would affect the market equilibrium price and quantity, as well as the output of individual (poultry) farmers, in the short run.

Overall performance was satisfactory. Some candidates seemed to have had an insufficient understanding of the meaning and implications of "a perfectly competitive market".

Question 2(c)(ii) – 5 Marks

This question required candidates to explain how a negative demand shock (due to bird flu) would affect the market equilibrium price and quantity, as well as the output of individual (poultry) farmers, in the long run.

Overall performance was satisfactory, though indeed far worse than part (c)(i). The poorer performance seemed to be due to candidates' failure to understand the features of a perfectly competitive market in the long run and the meaning of the phrase "Suppose the change in demand mentioned in part (c)(i) is *permanent*". Given that firms could have different cost structures in a perfectly competitive market (e.g., some could be more cost effective than others) and firms can exit the market in the long run, a permanently lowered market price (due to the bird flu) would cause the least cost competitive firms to leave the market. This would reduce the market supply and increase the market price in the long run, resulting in a higher output from the (price-taking) individual farmers.

Question 3(a) – 2 Marks

This question required candidates to define confidence interval.

Overall performance was unsatisfactory. Most candidates seemed to be unfamiliar with the term. Candidates are reminded that in order to perform well in a definition question, prior knowledge of the term/concept must be acquired through studying the relevant material before the exam.

Question 3(b) – 4 Marks

This question required candidates to compute the confidence interval for a population proportion (in this case, percentage of households claiming to own a made-in-Japan flat screen TV).

Overall performance was unsatisfactory. This was a standard calculations-based question, but most of the candidates who answered this question were not able to write the correct formula. This clearly reflected a lack of preparation and candidates' weak background in statistics.

Question 3(c) – 2 Marks

This question required candidates to interpret the confidence interval estimate calculated in part (b).

Overall performance was worse than part (b). Most of the candidates who answered the question simply put down the same definition used to answer part (a). However, “interpreting” a numerical answer usually requires answering “in context”, which most candidates failed to realise.

Question 3(d) – 1 Mark

This question asked candidates to state (with no explanation required) whether a lower confidence level (e.g., 90% instead of 95%) would result in a wider or narrower confidence interval.

Overall performance was satisfactory.

Question 4(a) – 3 Marks

This question asked candidates to explain the difference between a parameter and a sample statistic.

Overall performance was unsatisfactory. Candidates seemed to have no knowledge of the term. This once again reflected poor exam preparation and candidates’ weak background in statistics.

Question 4(b) – 2 Marks

This question required candidates to state two examples of a parameter.

Overall performance was unexpectedly unsatisfactory. Many candidates’ answers were irrelevant to the question.

Question 4(c) – 4 Marks

This question required candidates to compute the range and arithmetic average of a stock’s quarterly return.

Overall performance was satisfactory.

Question 4(b)(ii) – 4 Marks

This question asked candidates to define and compute the median and mode of the stock returns.

Overall performance was good.

Question 4(e) – 4 Marks

This question required candidates to describe the position and shape of the stock return distribution.

Overall performance was satisfactory. Some candidates with poor performance mostly failed to realise that the absolute values of the data in both the positive and negative regions are exactly the same, implying that the mean of the distribution is zero and its shape is symmetric.

Question 4(f) – 4 Marks

This question asked candidates to compute the standard deviation of the stock returns.

Overall performance was satisfactory. However, some candidates seemed to have no knowledge of the term at all.

Question 4(g) – 4 Marks

This question required candidates to explain how the conversion of nominal return into real return (by subtracting from the former the quarterly inflation rate) would affect the mean and standard deviation values.

Overall performance was unsatisfactory. The poor performance of some candidates was evidently due to a lack of sufficient understanding of the structure of the formulas for the two measures.

(III) Conclusion and Recommendation

Overall, the performance of the candidates was similar to the last cohort.

The key factors that held back the candidates' performance seemed to be a serious lack of preparation for the exam and a poor background in both economics and statistics.

As repeatedly suggested before, the most efficient and effective way to gain a basic understanding of the whole syllabus and the types of questions that candidates might face in the examination is to go through the reference readings assigned for this module. It would also be very helpful if candidates could spend time going through past exam questions to familiarise themselves with how the issues could be examined. For candidates with weak backgrounds in economics and statistics, consulting relevant reference books on both subjects should be an integral part of the preparation process.